

Stack Overflow Developer Survey 2024

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OUTLINE



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EXECUTIVE SUMMARY



- **Technology Trends**
 - JavaScript, HTML/CSS, and Python remain the most used languages; Python is now the most desired, and Rust is the most admired.
- **Learning & Career Development**
 - Developers favor online learning (82%) over formal education (49%), with technical documentation and Stack Overflow as top resources.
 - The developer population is aging (39% are 35+), but many are still early in their careers and continue to learn actively.
- **AI Adoption & Perceptions**
 - 76% are using or planning to use AI tools, with mostly positive attitudes.
 - Trust in AI is mixed—only 43% trust its accuracy, and 45% doubt its ability with complex tasks.
 - Most developers (70%) don't see AI as a job threat.
- **Workplace Experience**
 - Technical debt is the top frustration (62%), followed by build and deployment complexity.
 - Salaries have dropped across several roles, but most developers (68%) remain content, finding satisfaction in improving code and systems.



INTRODUCTION



- **Purpose:** To reveal key trends, technologies, challenges, and attitudes shaping the global developer community in 2024.
- **Target Audience:** Developers, tech leads, hiring managers, and companies building tools for developers.
- **Value:** Provides data-driven insights on popular tools, learning habits, workplace frustrations, and AI adoption—helping stakeholders understand and support developers more effectively.



METHODOLOGY



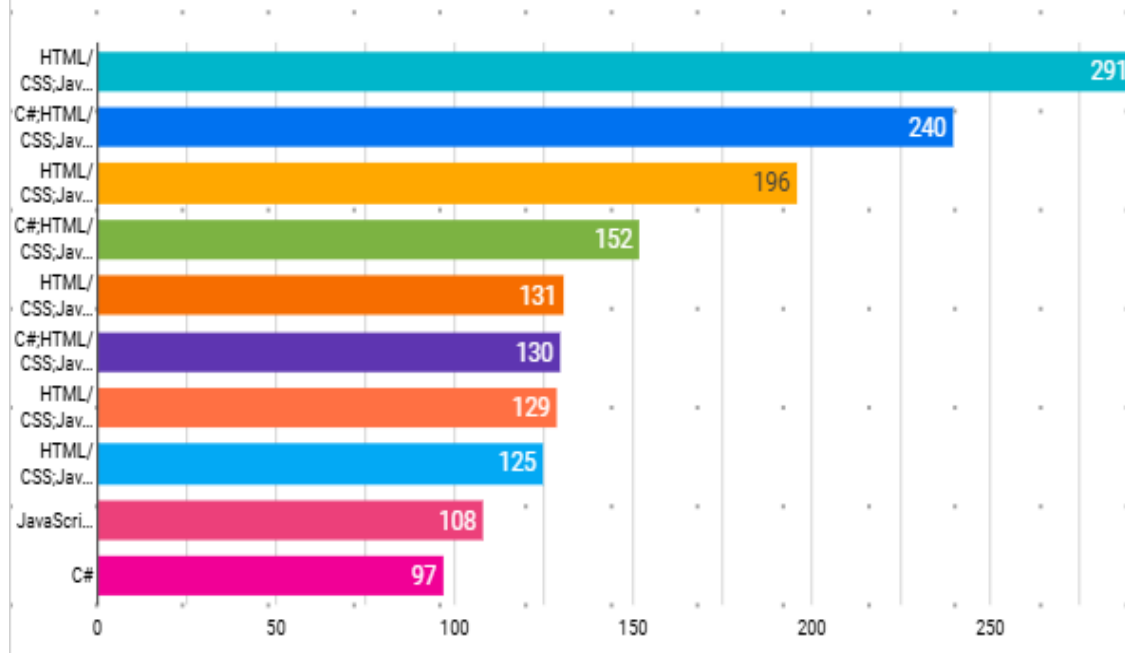
- Conducted in May 2024, targeting developers and technologists worldwide.
- Covered topics like technology use, learning methods, job roles, salaries, and AI perceptions.
- Included multiple-choice and opinion questions, with updated categories like embedded tech and AI tools.
- Collected demographic data (e.g. age, education, experience).
- Analyzed trends in language popularity, developer frustrations, and career satisfaction.
- AI adoption was a key focus, alongside persistent challenges like technical debt.
- Results reflect the Stack Overflow user base, not the entire global developer community.



PROGRAMMING LANGUAGE TRENDS

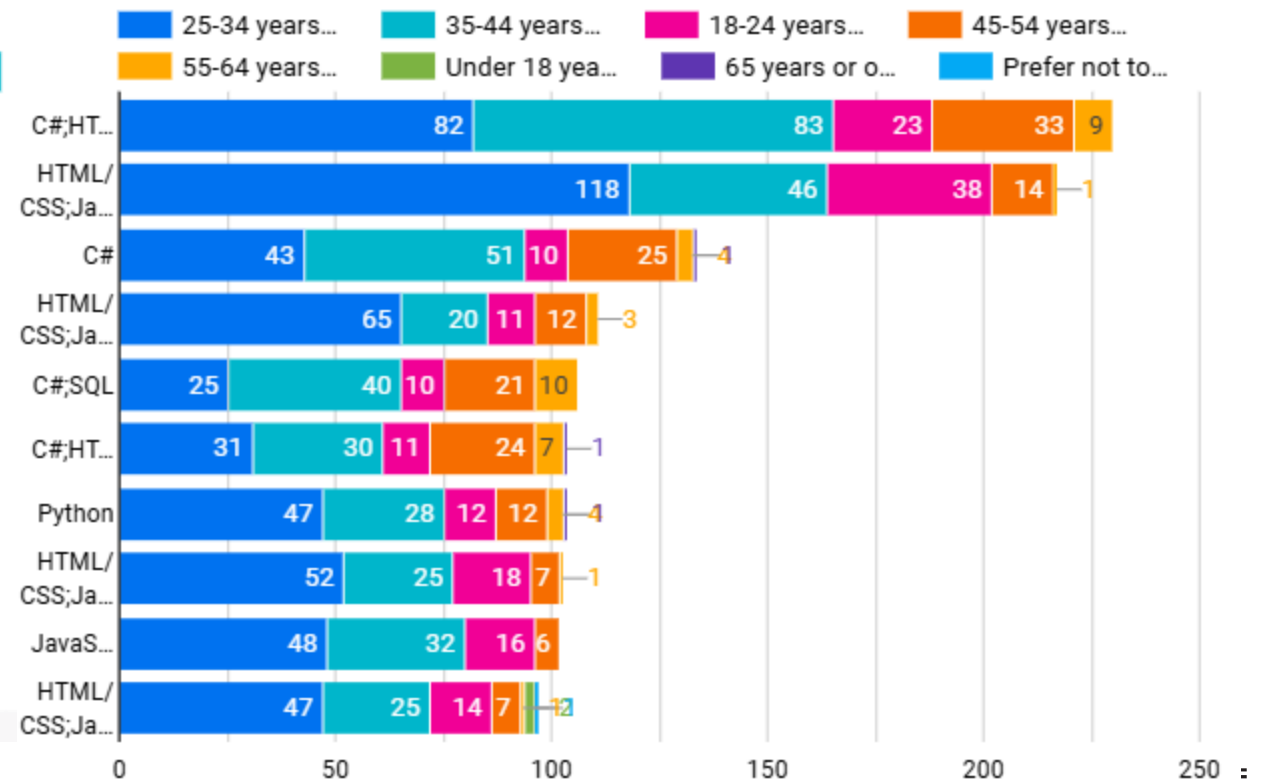
Current Year

Languages Used By Respondants



Next Year

Record Count by LanguageWantToWorkWith and Age



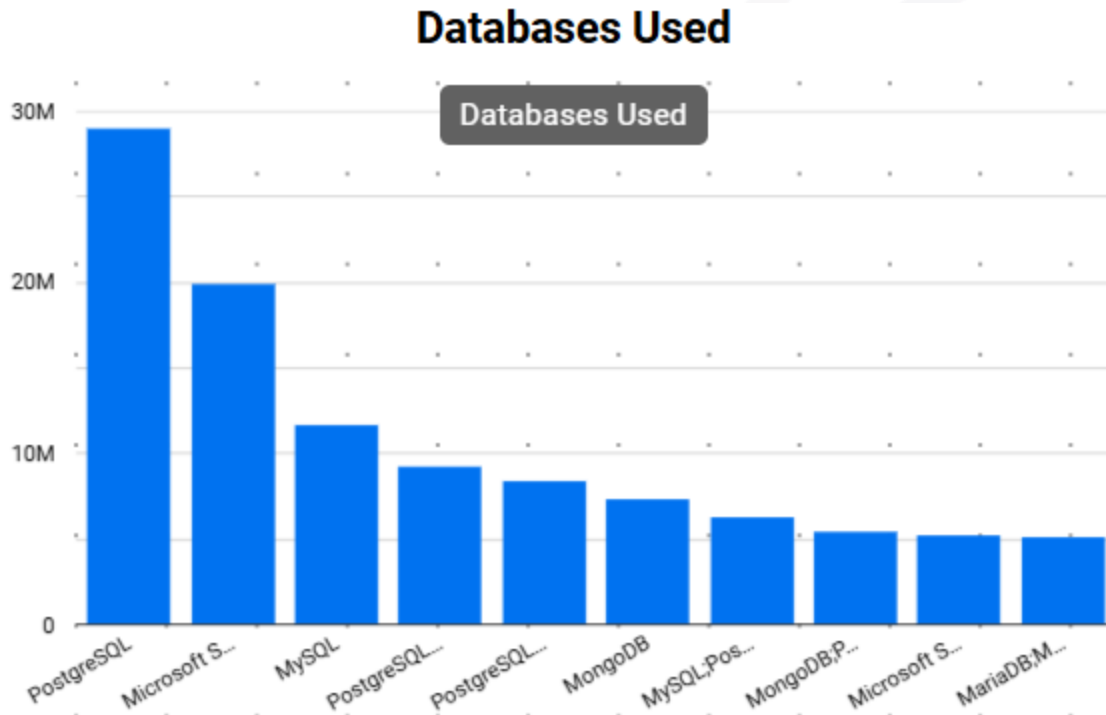
PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

- **JavaScript** remains the most widely used programming language.
- **Rust** is the most admired language (83% want to keep using it).
- **Python and SQL** are highly desired and widely used.
- **PostgreSQL** is the most popular database (used by 49% of developers).
- **For Developers:**
 - Maintain strong skills in JavaScript, Python, and SQL.
 - Learn Rust for performance-critical and systems programming roles.
- **For Organizations:**
 - Adopt tools like Rust and PostgreSQL for performance and scalability.
 - Support developer-preferred languages to boost satisfaction and retention.

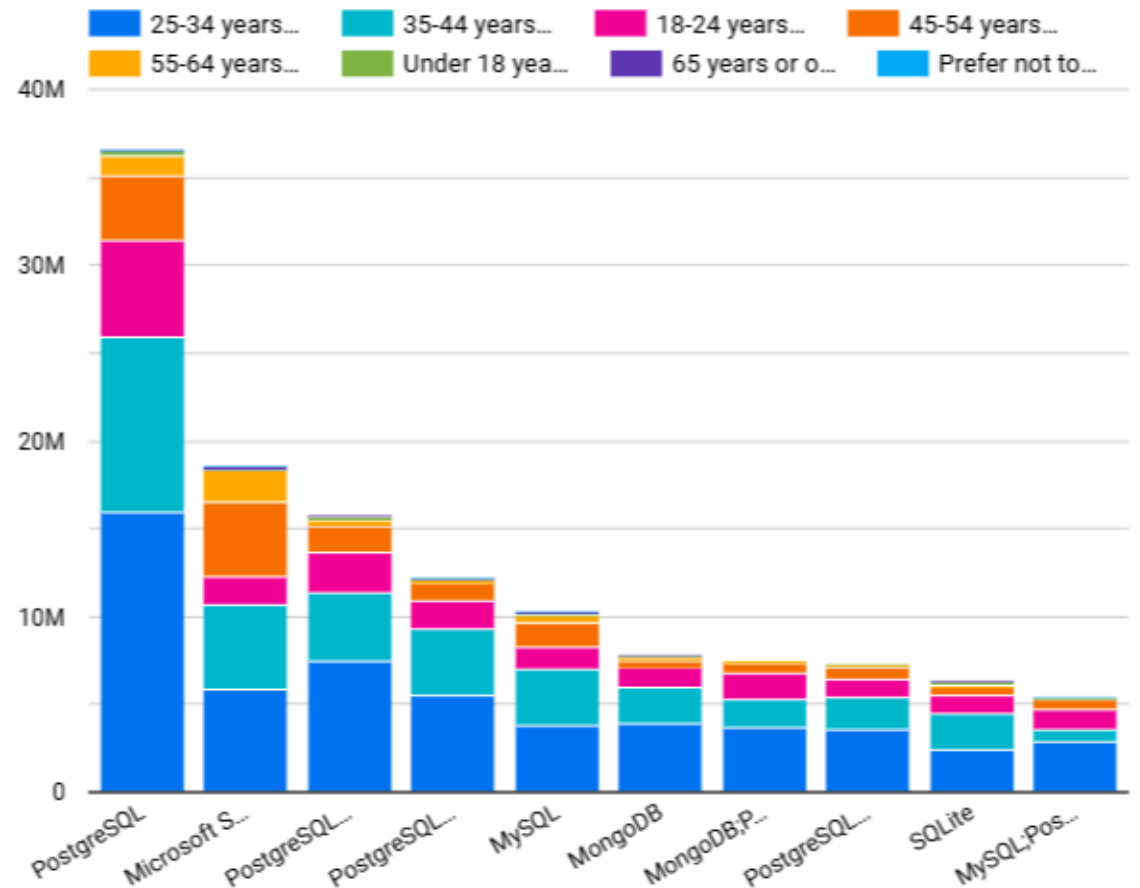


DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

- **PostgreSQL** is the most popular database, used by 49% of developers (up from 33% in 2018).
- **MySQL** has seen a decline in usage from 59% in 2018.
- **For Developers** Learning PostgreSQL is increasingly valuable for backend and data engineering roles.
- **For Organizations** Adopting PostgreSQL offers cost savings, scalability, and strong community support.

Implications

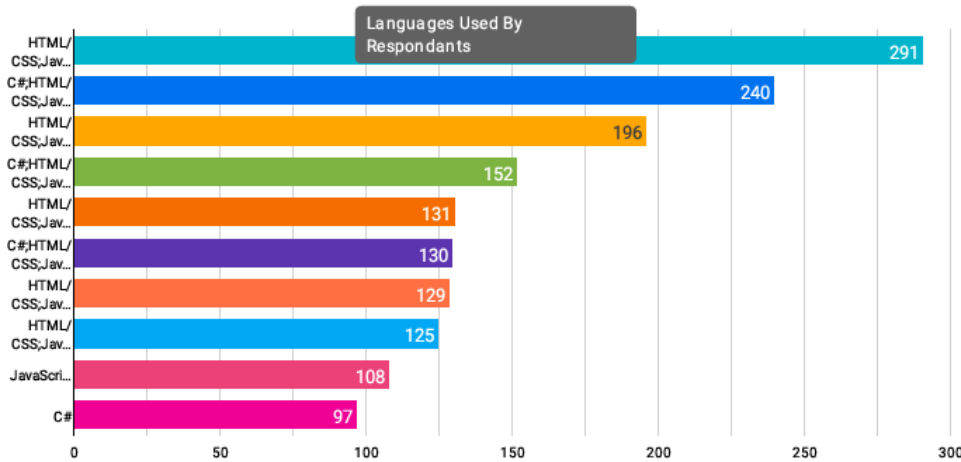


DASHBOARD

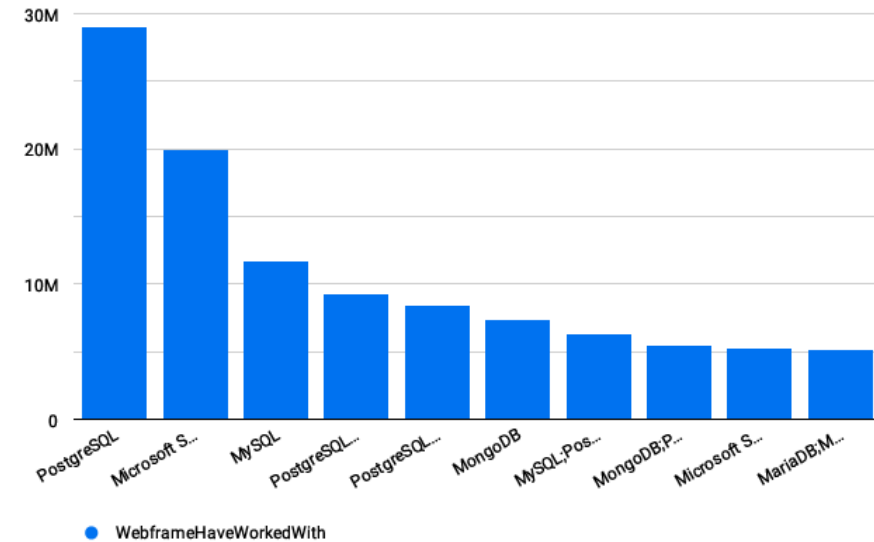


DASHBOARD TAB 1

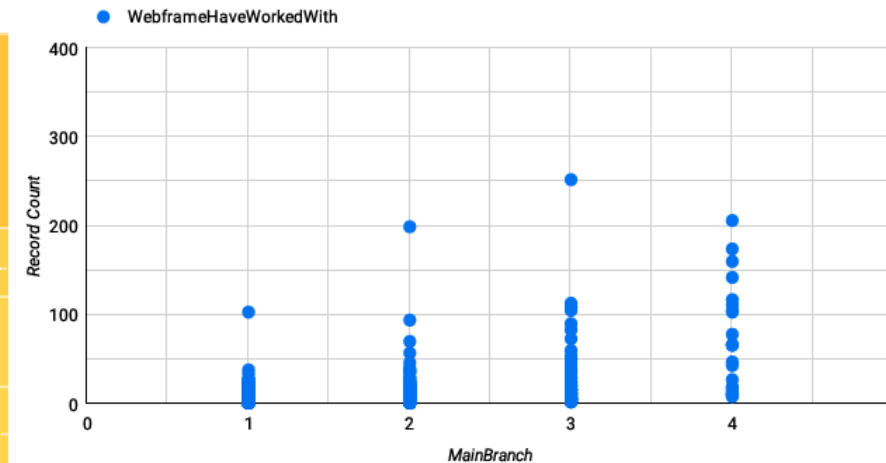
Languages Used By Respondants



Databases Used

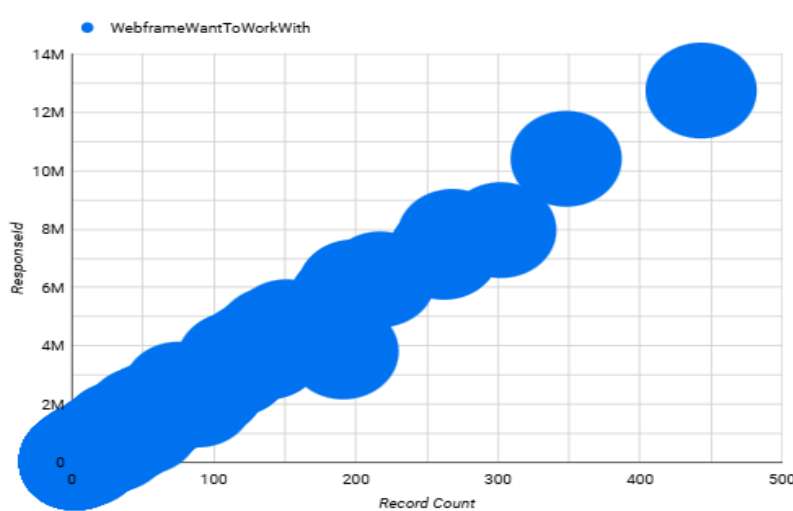
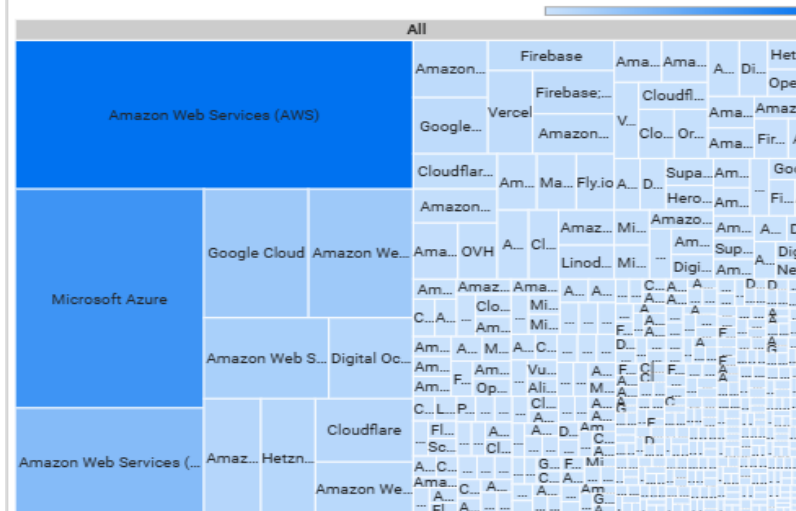
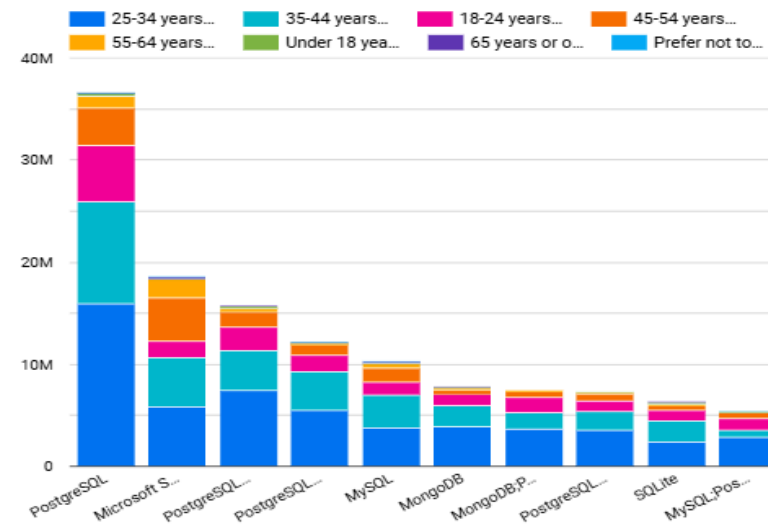
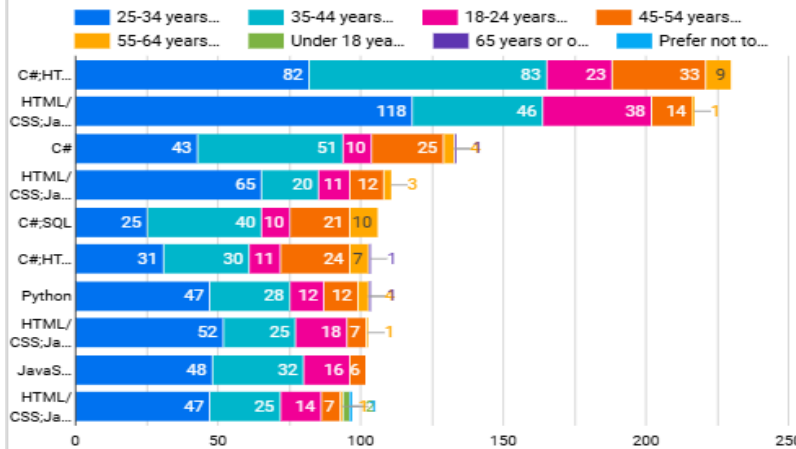


Platforms Used

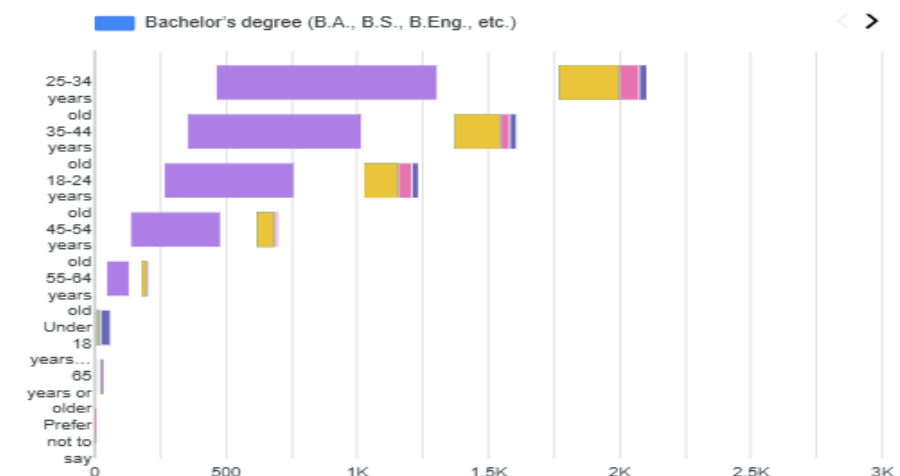
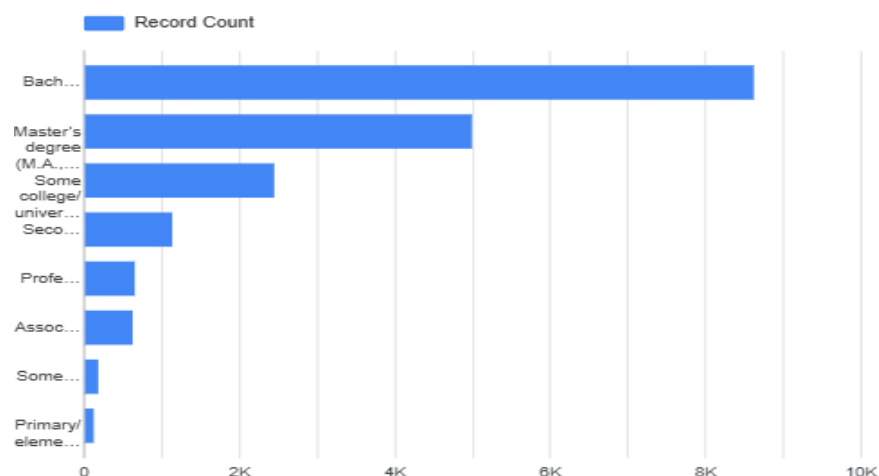
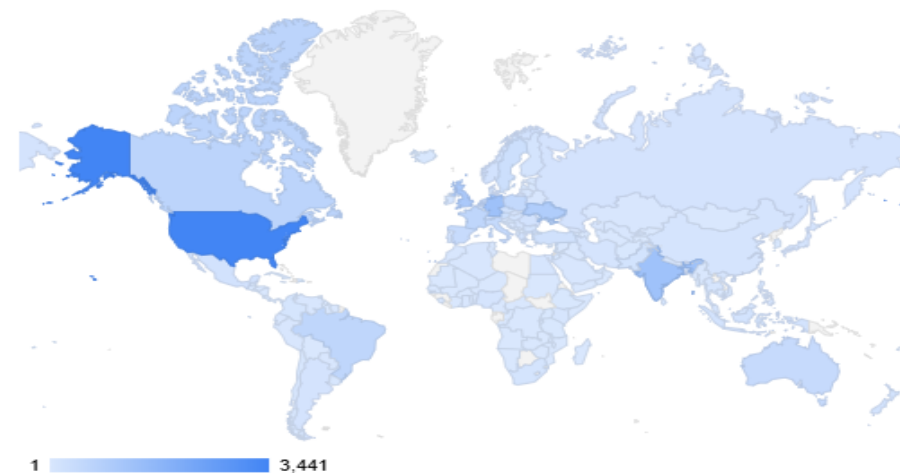
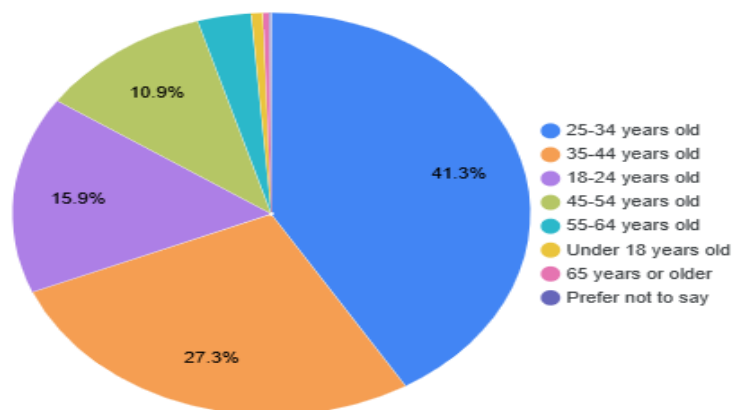


DASHBOARD TAB 2

Record Count by LanguageWantToWorkWith and Age



DASHBOARD TAB 3



DISCUSSION



- Developer Demographics: 36.5% of developers are aged 25-34, 66% have a BA/BS or MA/MS, and 82% learn to code through online resources.
- Technology Trends: JavaScript, Python, and SQL remain popular, with PostgreSQL gaining significant traction as the top database.
- AI Adoption: 76% of developers use or plan to use AI tools, with benefits like increased productivity, though concerns about accuracy and misinformation persist.
- Work Environment: 84% of developers are employed full-time, with 42% working in hybrid models and 62% visiting the office weekly.



OVERALL FINDINGS & IMPLICATIONS

- **Demographics:** The developer workforce is largely young (36.5% are aged 25-34), with a strong focus on self-learning through online resources (82%).
- **Technology Preferences:** JavaScript, Python, and SQL dominate, with PostgreSQL emerging as the leading database. Rust is highly admired.
- **AI Usage:** Developers are increasingly using AI tools (76%), appreciating productivity benefits but concerned about accuracy and ethical issues.
- **Work Trends:** Hybrid work models are most common (42%), with many developers returning to the office part-time.
- **For Developers:** Keeping up with popular languages like JavaScript, Python, and SQL, and exploring emerging tools like Rust and PostgreSQL, can boost career prospects.
- **For Organizations:** Embracing hybrid work models and AI tools can enhance productivity, while addressing concerns around ethical AI usage and maintaining trust.



CONCLUSION



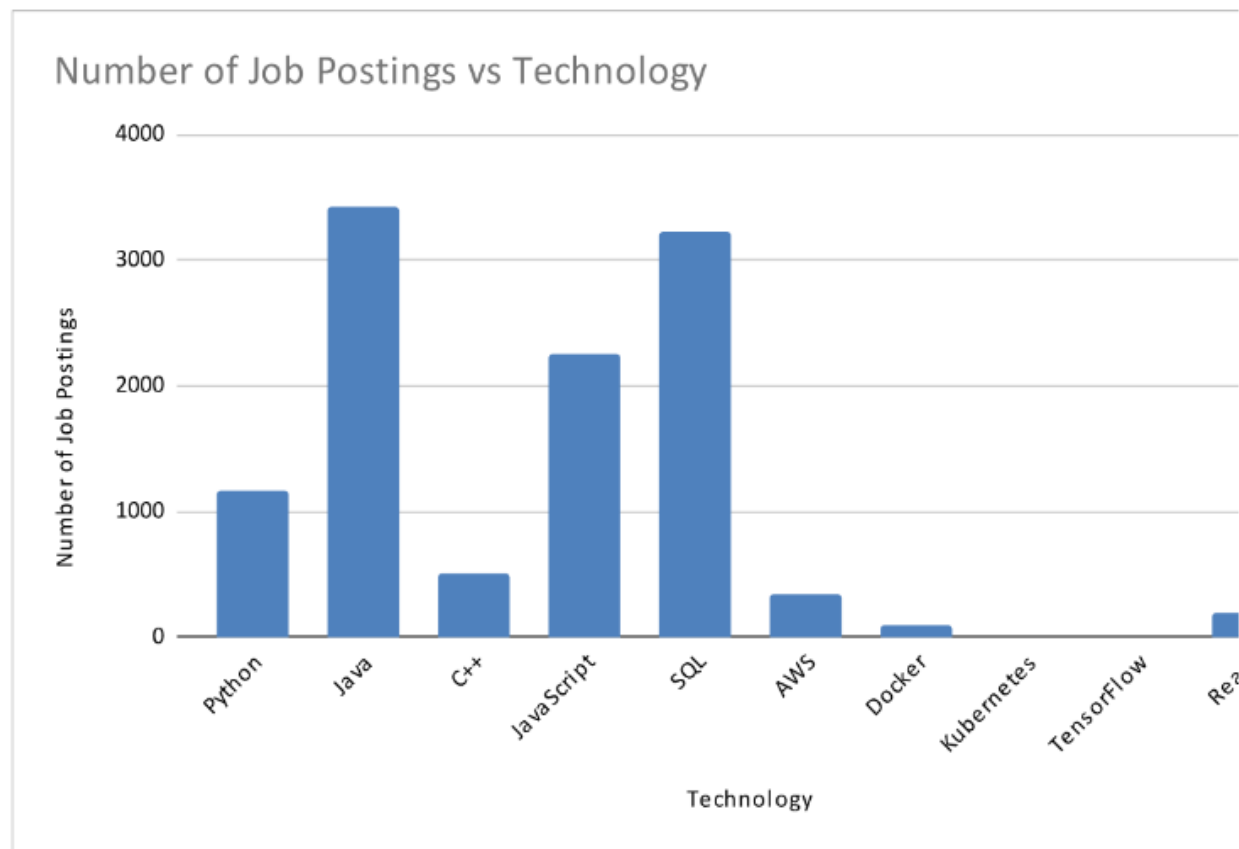
- **Continued Demand for Core Technologies:** JavaScript, Python, and SQL remain essential, with PostgreSQL gaining strong adoption.
- **AI Integration is Increasing:** Developers are embracing AI tools, but ethical concerns and accuracy issues need attention.
- **Work Preferences are Shifting:** Hybrid work is the preferred model, with flexibility being key to developer satisfaction.
- **Self-Learning is Crucial:** Online resources are the primary way developers acquire new skills, highlighting the need for continuous learning.



APPENDIX



JOB POSTINGS



POPULAR LANGUAGES

